4 Decision Making for a Single Sample

The **SPACE SHUTTLE CHALLENGER DISASTER** was a tragic event caused by the failure to heed engineering warnings regarding freezing temperatures and critical equipment seals.

- **Date and Event**: On January 28, 1986, the **Challenger** disintegrated 73 seconds after liftoff, resulting in the deaths of all seven astronauts on board.
- **Engineering Warning**: Roger M. Boisjoly, a lead engineer, urgently recommended delaying the launch based on data showing that freezing temperatures could cause critical seals in the solid rocket boosters to fail.

- **Management Decision**: NASA management ignored these warnings, effectively testing the engineer's hypothesis under real-world conditions.
- **Outcome**: The **Challenger** broke apart and fell into the sea, leaving no survivors after the predicted seal failure occurred.

4-1 STATISTICAL INFERENCE

- **Statistical inference** involves methods for making decisions or drawing conclusions about a **population**.
- These methods rely on information gathered from a **random sample**.
- **Figure 4-1** illustrates how the **population** and **random sample** relate to one another.
- This study focuses on how these statistical methods facilitate effective decision-making.

Statistical inference is primarily divided into two main branches:

- **Parameter estimation**: This involves calculating a specific value (like a mean or average) to represent a population. For instance, an engineer might estimate the average tensile strength of car chassis components to account for natural variations in manufacturing and materials.
- **Hypothesis testing**: This involves using data to evaluate specific claims or theories about a population.

In practice, using an estimator allows professionals to determine the **precision** of their data and understand the reliability of their results.